# EXPERIMENT: 7

## A PYTHON PROGRAM TO IMPLEMENT DECISION TREE

## Aim:

To implement a Decision Tree classifier for classification.

### Procedure:

1. Import necessary libraries.
2. Load dataset (Iris dataset).
3. Split data into training and testing sets.
4. Train Decision Tree model.
5. Perform prediction on test data.
6. Evaluate accuracy.

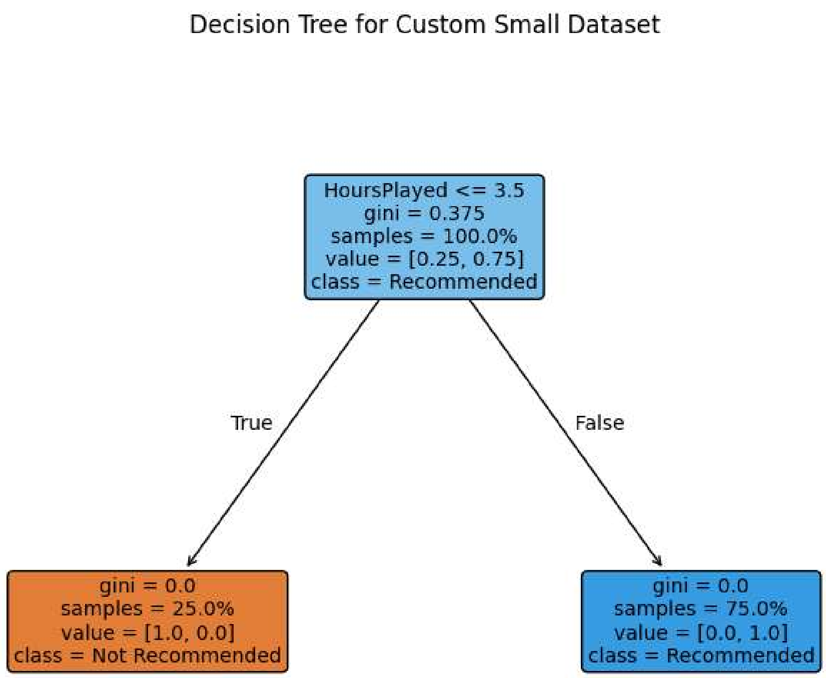
## CODE:

```
import pandas as pd

from sklearn.tree import DecisionTreeClassifier, plot_tree

from sklearn.model_selection import train_test_split

from sklearn.metrics import accuracy_score

import matplotlib.pyplot as plt



data_custom = {

   'HoursPlayed': [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],

   'EnjoymentScore': [3, 4, 5, 6, 7, 7, 8, 8, 9, 10],

   'Recommended':    [0, 0, 0, 1, 1, 1, 1, 1, 1, 1]

}
```

```
df_custom = pd.DataFrame(data_custom)

X_custom = df_custom[['HoursPlayed', 'EnjoymentScore']]

y_custom = df_custom['Recommended']

X_custom_train, X_custom_test, y_custom_train, y_custom_test =
train_test_split(

   X_custom, y_custom, test_size=0.2, random_state=42

)

model_custom = DecisionTreeClassifier(random_state=42)

model_custom.fit(X_custom_train, y_custom_train)

plt.figure(figsize=(10, 7))

plot_tree(

    model_custom,

    feature_names=X_custom.columns,

    class_names=['Not Recommended', 'Recommended'],

    filled=True,

    rounded=True,

    proportion=True,

    fontsize=10

)

plt.title('Decision Tree for Custom Small Dataset')

plt.show()
```

```
y_custom_pred = model_custom.predict(X_custom_test)

print("Accuracy for Custom Dataset:",
accuracy_score(y_custom_test, y_custom_pred))
```

**OUTPUT:**




## RESULT:

The Decision Tree model classified the data correctly with satisfactory accuracy.